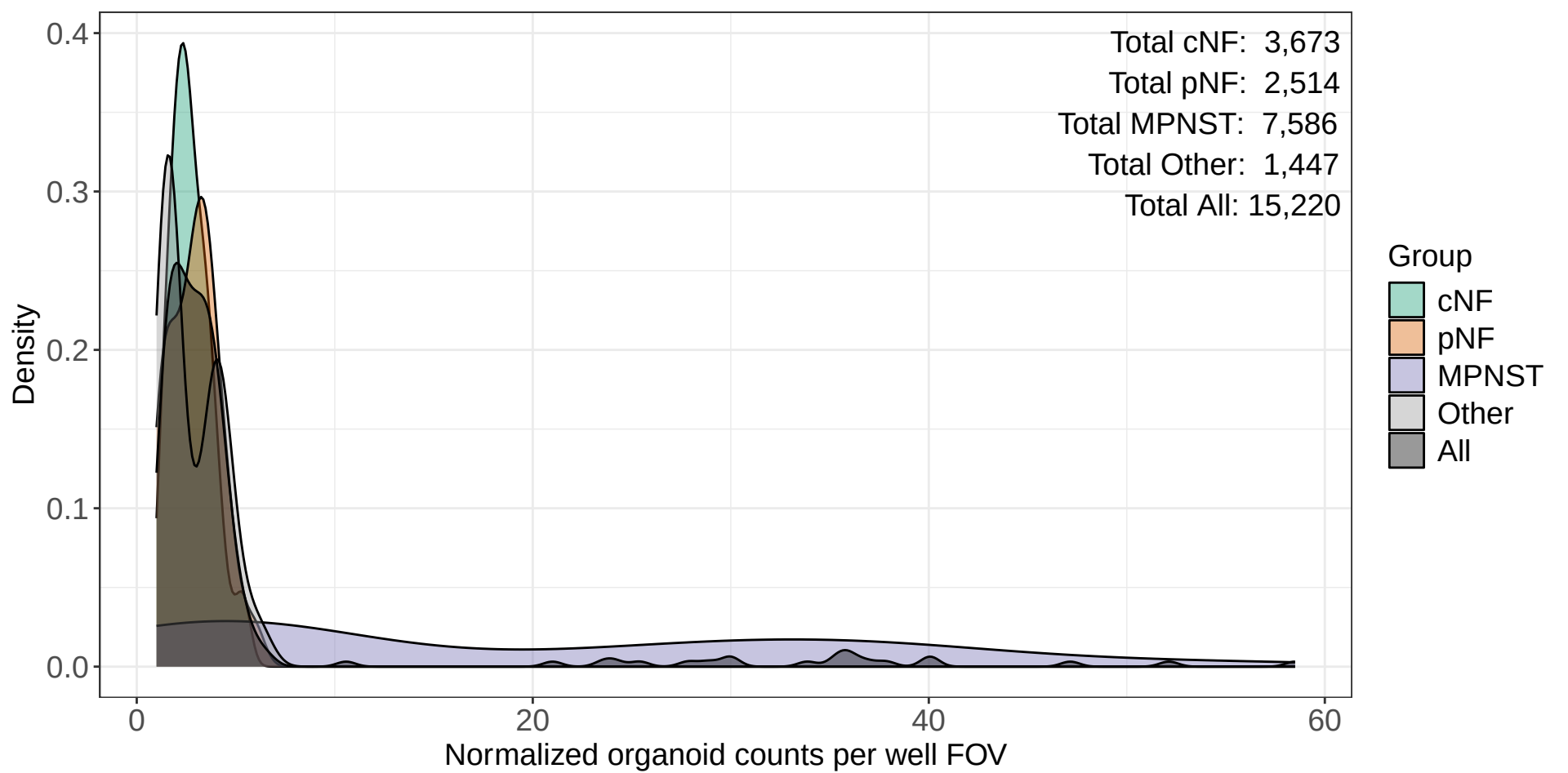
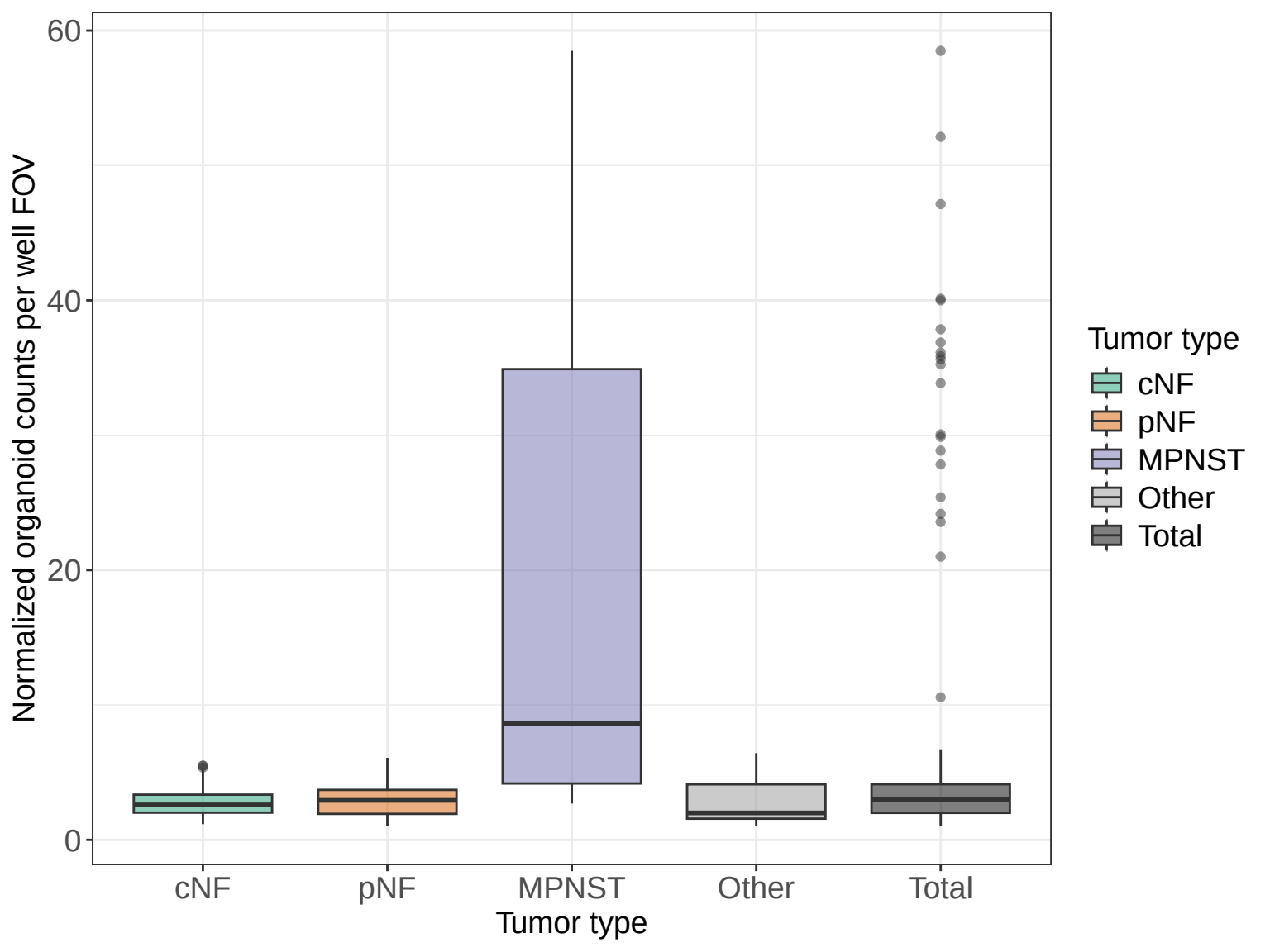
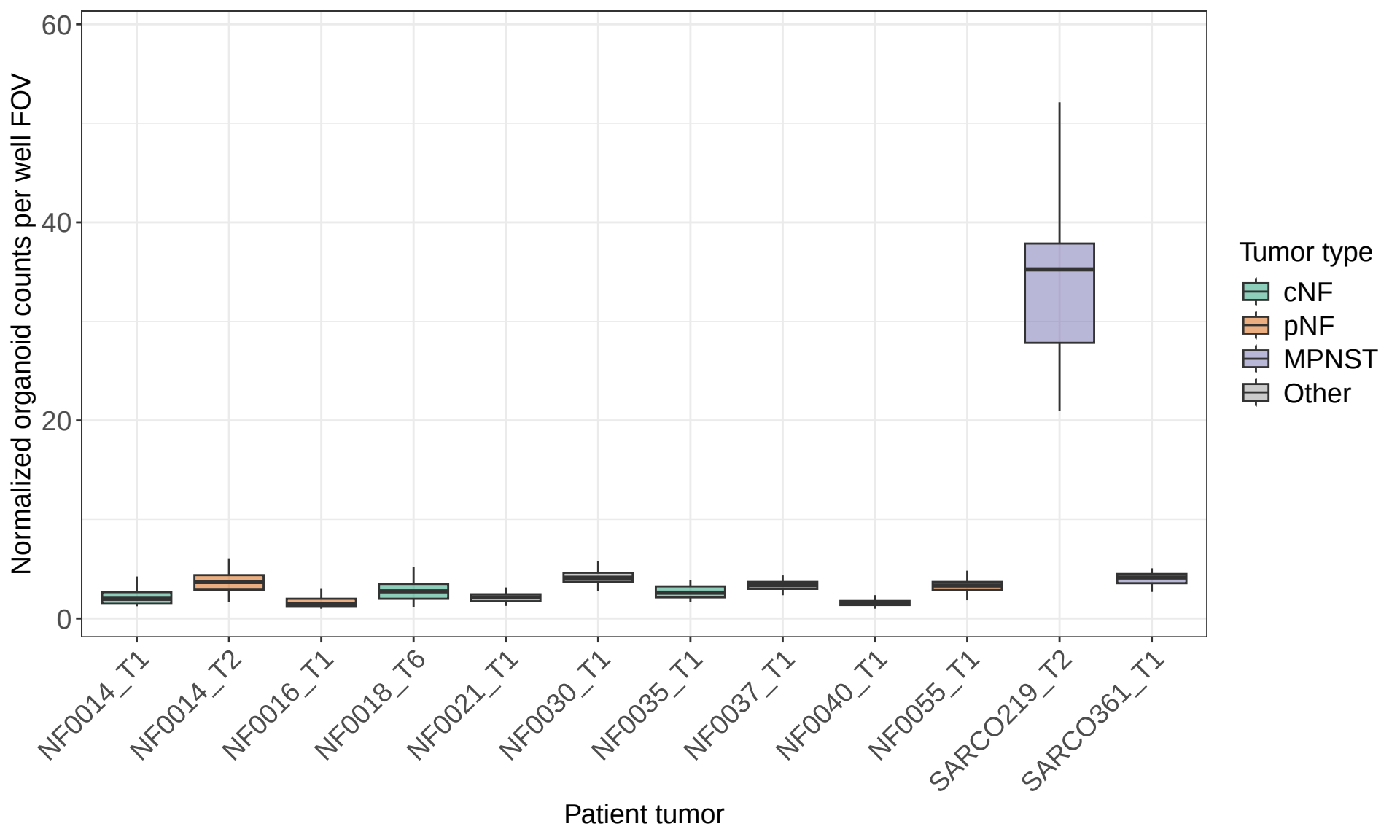
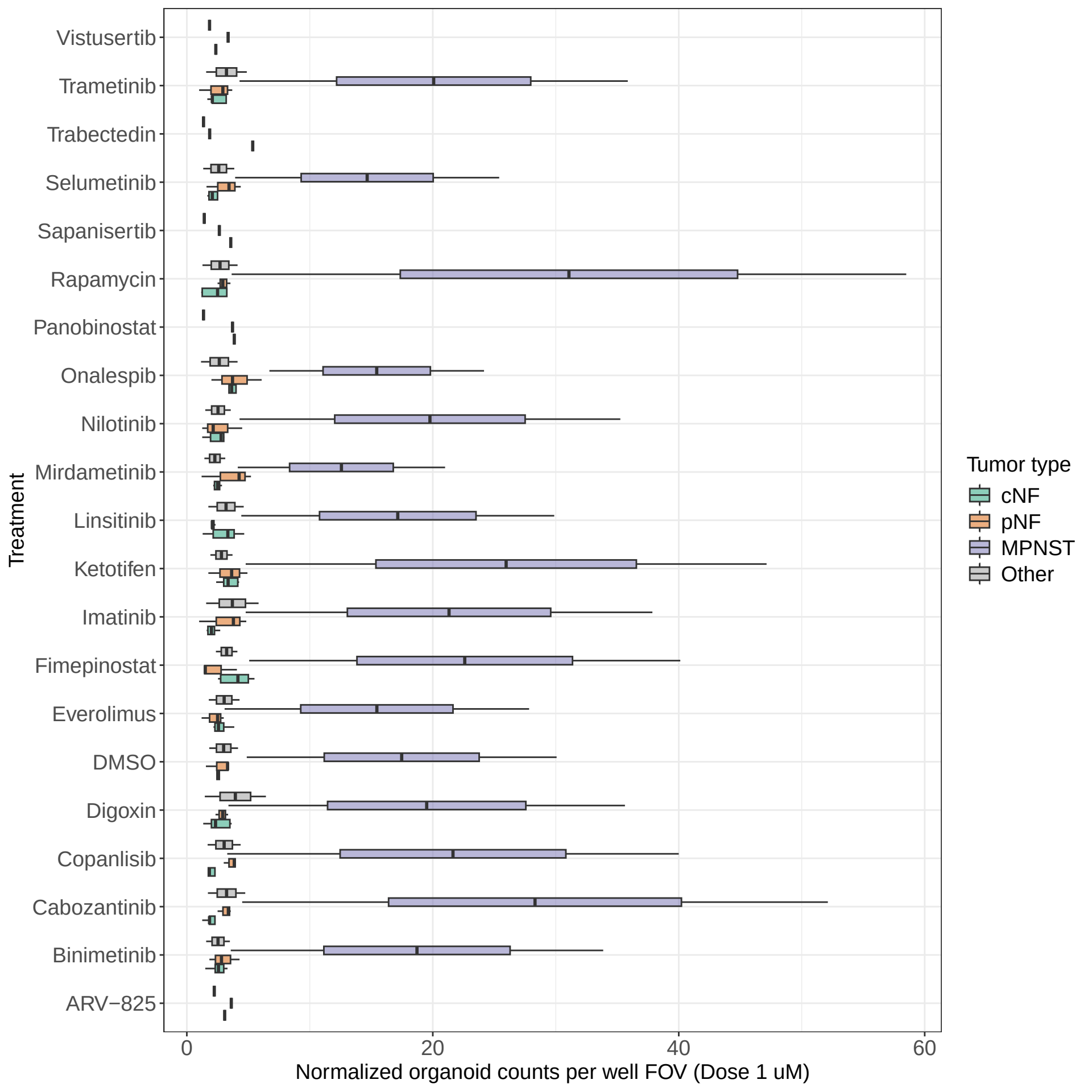
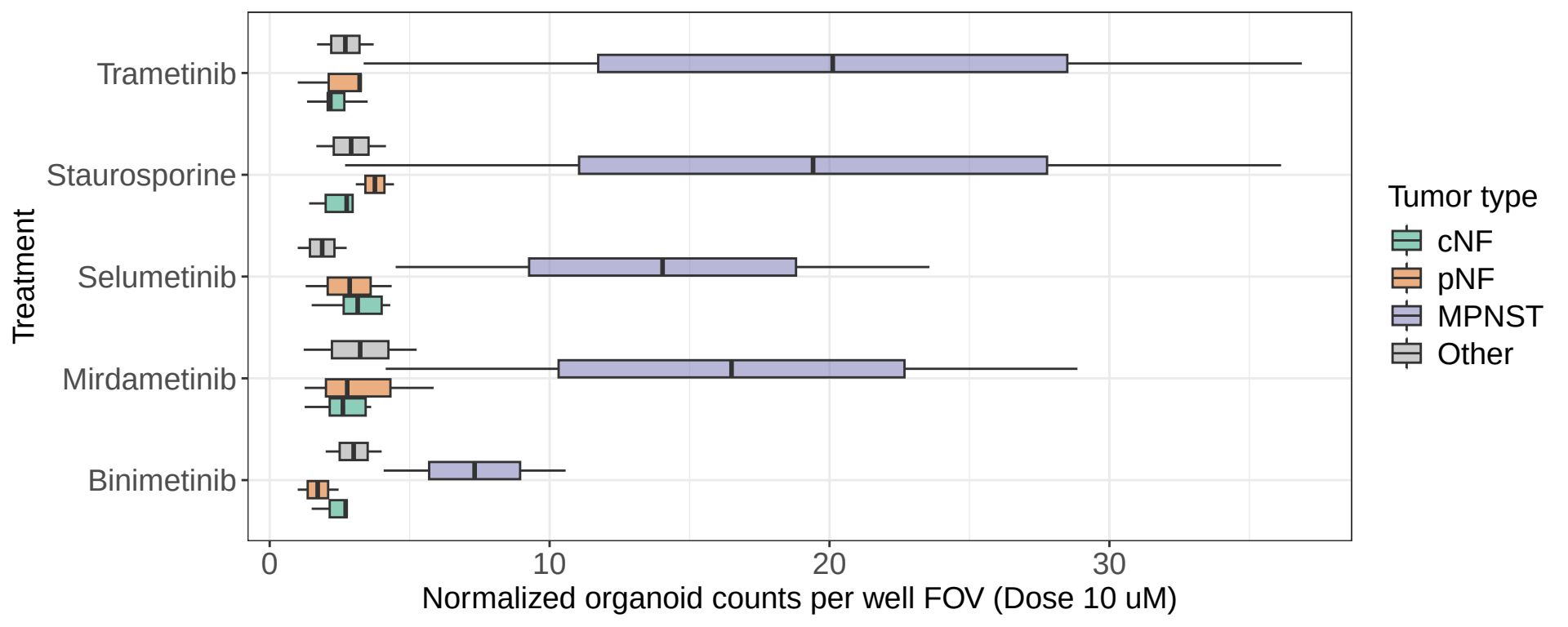
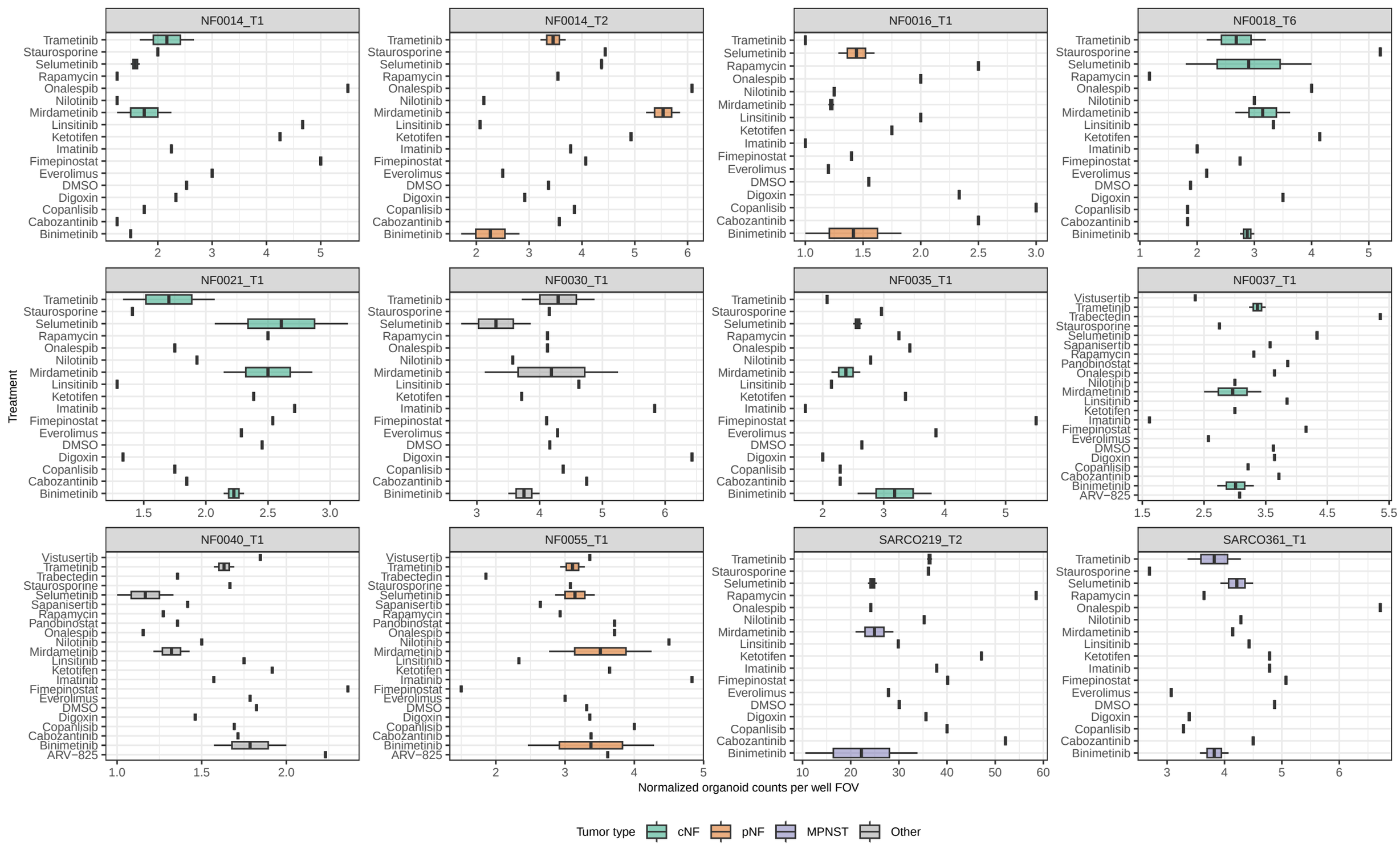

Total cNF: 3,673
Total pNF: 2,514
Total MPNST: 7,586
Total Other: 1,447
Total All: 15,220
Group
cNF
pNF
MPNST
Other
All
Density
0.4
0.3
0.2
0.1
0.0
0
20
40
60
Normalized organoid counts per well FOV

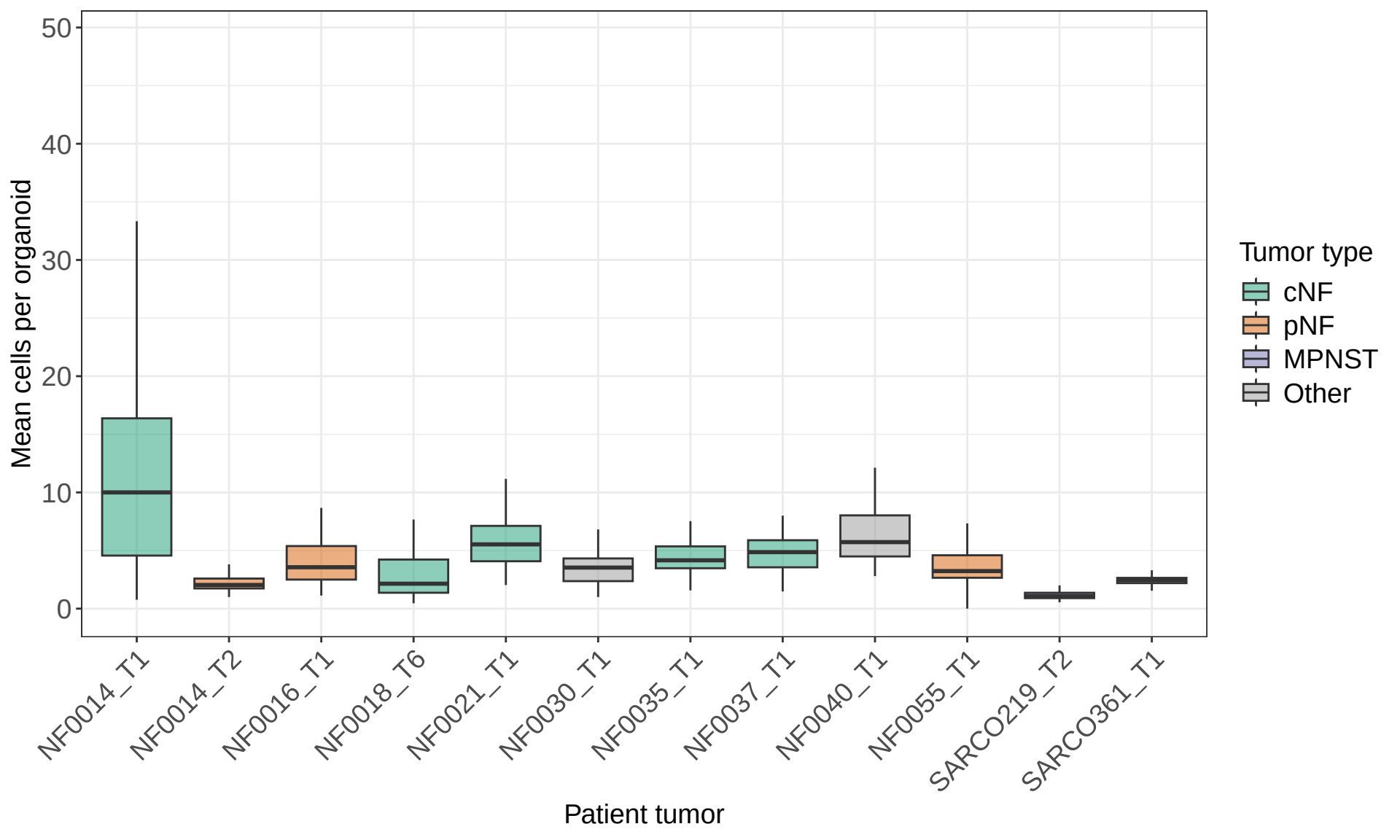

60
40
20
0
Normalized organoid counts per well FOV
cNF
pNF
MPNST
Other
Total
Tumor type
Tumor type
cNF
pNF
MPNST
Other
Total

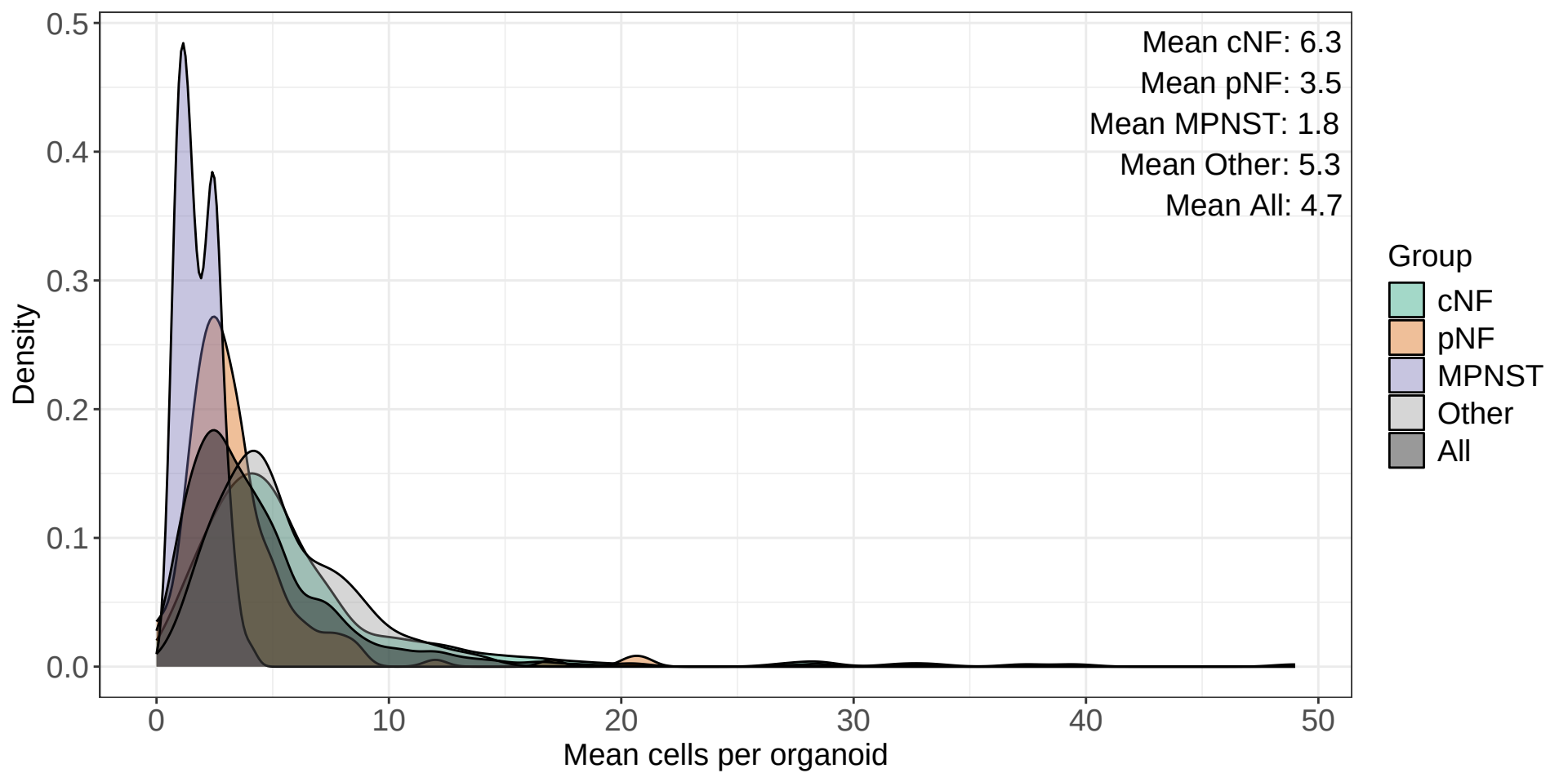

Normalized organoid counts per well FOV
60
40
20
0
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Patient tumor
Tumor type
cNF
pNF
MPNST
Other

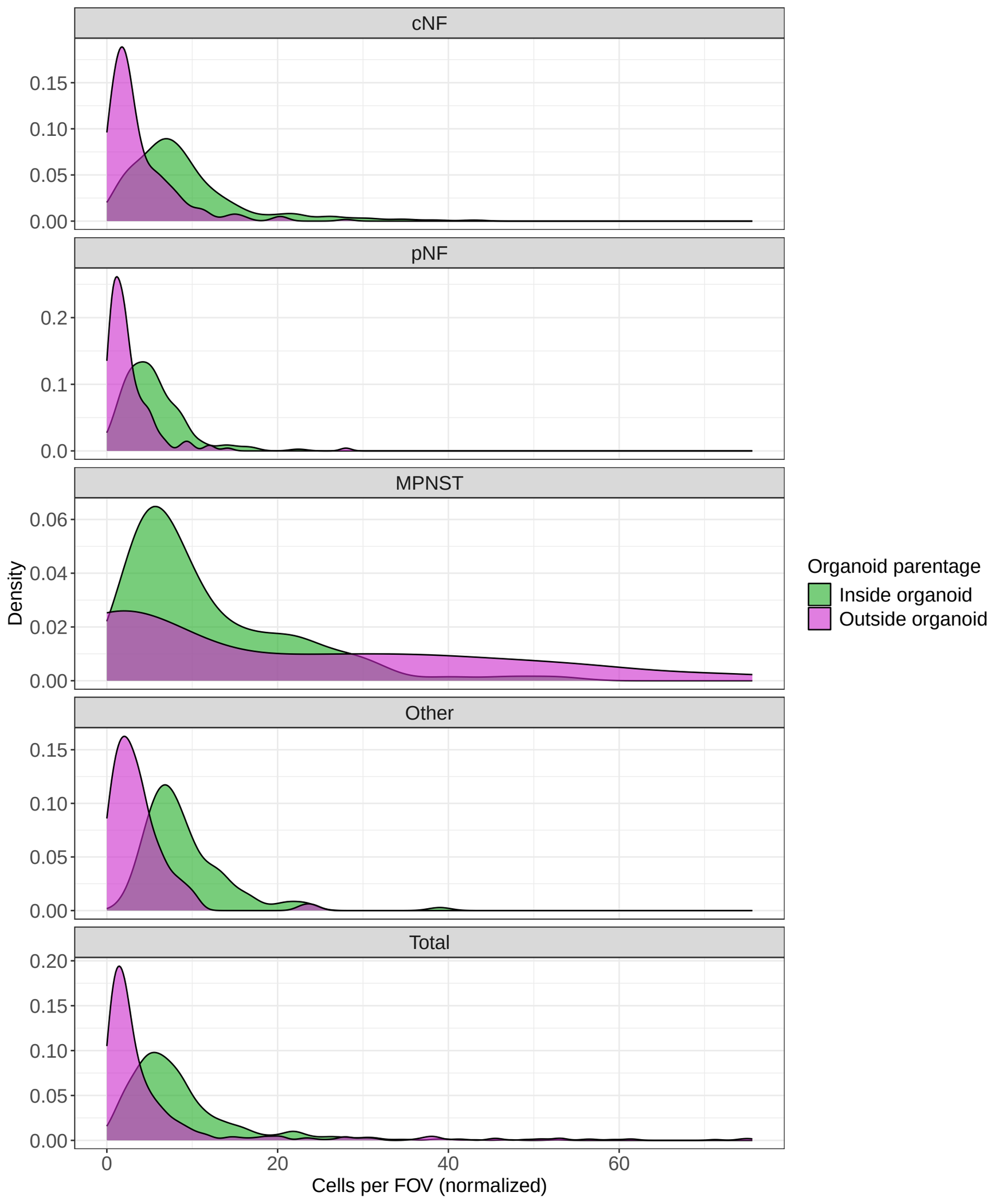

Vistusertib
Trametinib
Trabectedin
Selumetinib
Sapanisertib
Rapamycin
Panobinostat
Onalespib
Nilotinib
Mirdametinib
Linsitinib
Ketotifen
Imatinib
Fimepinostat
Everolimus
DMSO
Digoxin
Copanlisib
Cabozantinib
Binimetinib
ARV−825
Treatment
0
20
40
60
Normalized organoid counts per well FOV (Dose 1 uM)
Tumor type
cNF
pNF
MPNST
Other

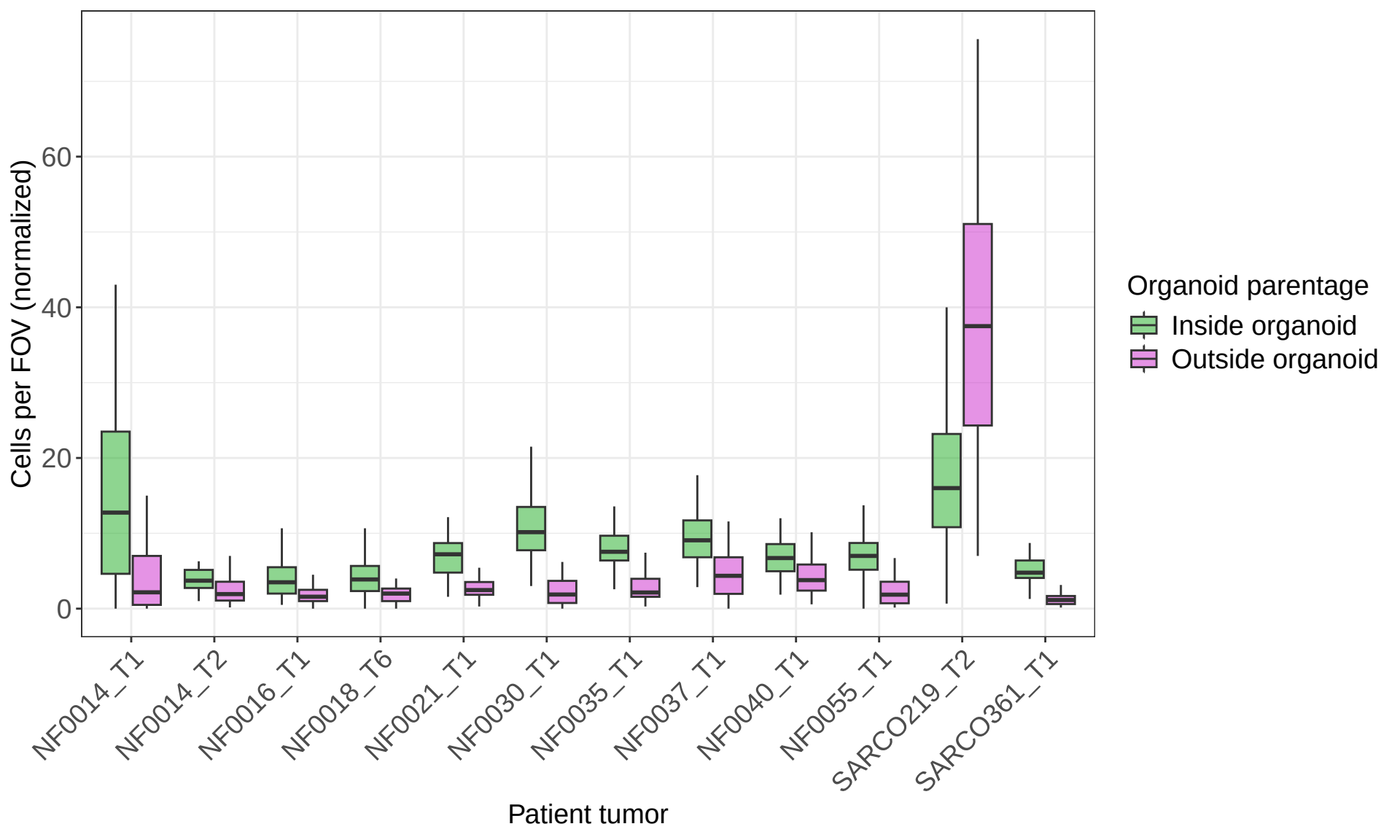

Trametinib
Staurosporine
Selumetinib
Mirdametinib
Binimetinib
Treatment
0
10
20
30
Normalized organoid counts per well FOV (Dose 10 uM)
Tumor type
cNF
pNF
MPNST
Other

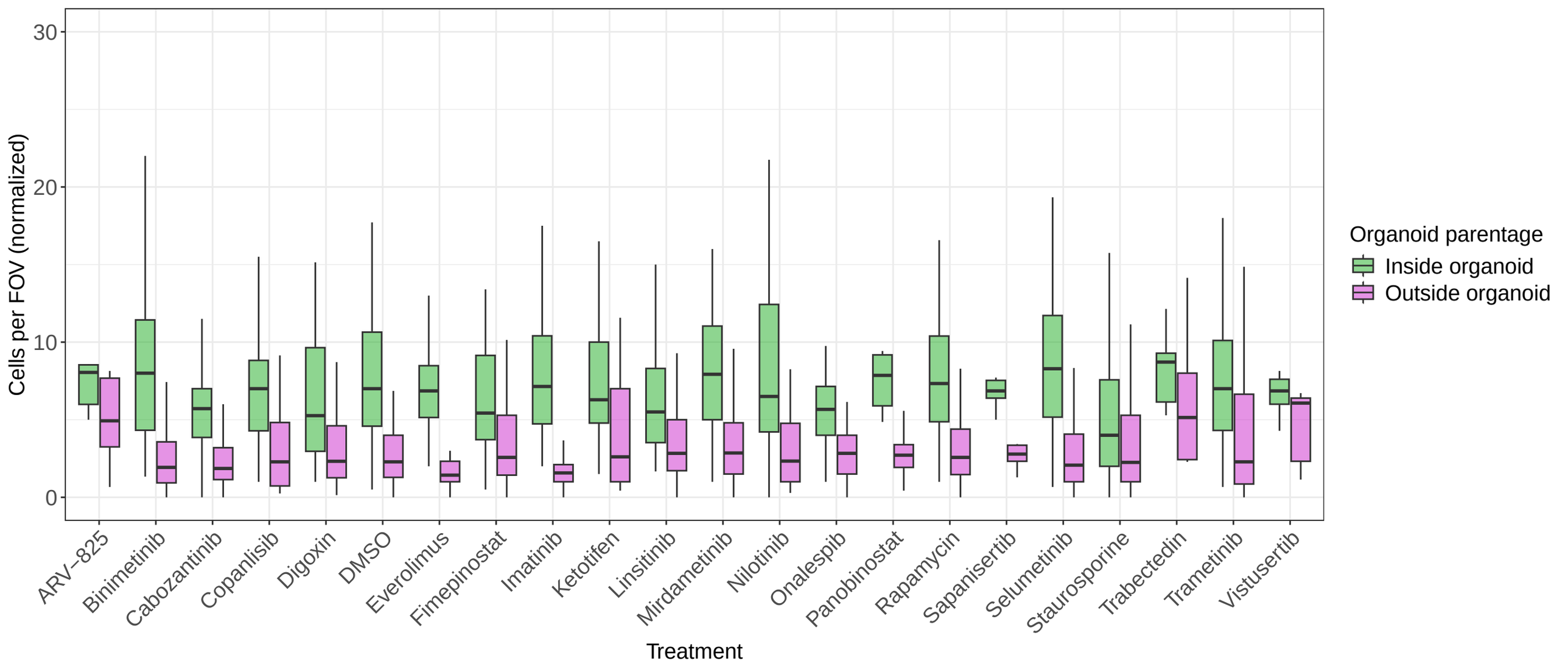

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Trametinib
Staurosporine
Selumetinib
Rapamycin
Onalespib
Nilotinib
Mirdametinib
Linsitinib
Ketotifen
Imatinib
Fimepinostat
Everolimus
DMSO
Digoxin
Copanlisib
Cabozantinib
Binimetinib
Vistusertib
Trabectedin
Sapanisertib
Panobinostat
ARV−825
Treatment
Normalized organoid counts per well FOV
Tumor type
cNF
pNF
MPNST
Other

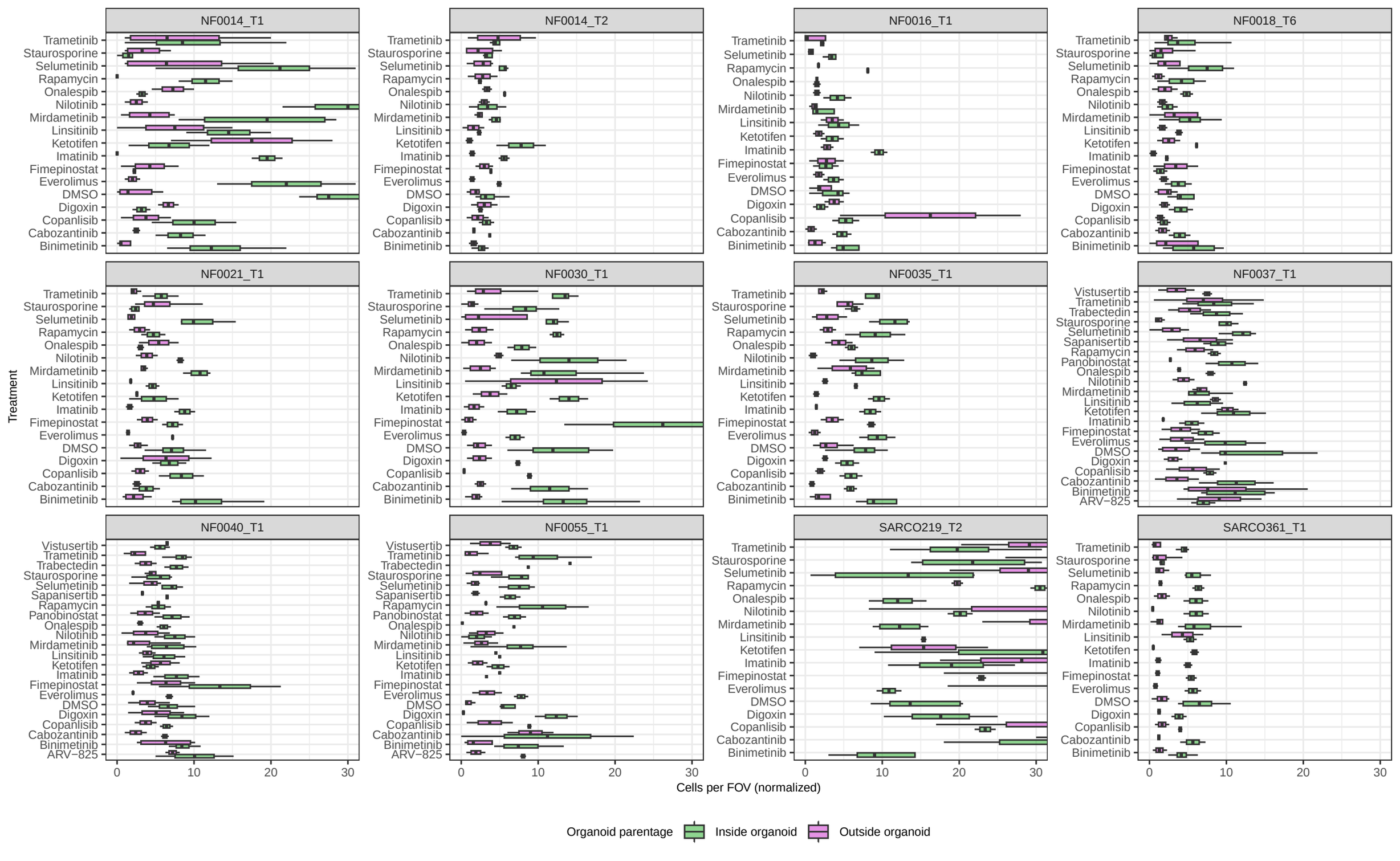

Mean cells per organoid
50
40
30
20
10
0
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Patient tumor
Tumor type
cNF
pNF
MPNST
Other


Mean cNF: 6.3
Mean pNF: 3.5
Mean MPNST: 1.8
Mean Other: 5.3
Mean All: 4.7
Group
cNF
pNF
MPNST
Other
All
Density
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
Mean cells per organoid


cNF
pNF
MPNST
Other
Total
Density
0.15
0.10
0.05
0.00
0.2
0.1
0.0
0.06
0.04
0.02
0.20
0
20
40
60
Cells per FOV (normalized)
Organoid parentage
Inside organoid
Outside organoid


Cells per FOV (normalized)
0
20
40
60
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Patient tumor
Organoid parentage
Inside organoid
Outside organoid


Cells per FOV (normalized)
30
20
10
0
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Treatment
Organoid parentage
Inside organoid
Outside organoid


NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Trametinib
Staurosporine
Selumetinib
Rapamycin
Onalespib
Nilotinib
Mirdametinib
Linsitinib
Ketotifen
Imatinib
Fimepinostat
Everolimus
DMSO
Digoxin
Copanlisib
Cabozantinib
Binimetinib
Vistusertib
Trabectedin
Sapanisertib
Panobinostat
ARV−825
Treatment
0
10
20
30
Cells per FOV (normalized)
Organoid parentage
Inside organoid
Outside organoid

cNF
pNF
MPNST
Other
Vistusertib
Trametinib
Trabectedin
Staurosporine
Selumetinib
Sapanisertib
Rapamycin
Panobinostat
Onalespib
Nilotinib
Mirdametinib
Linsitinib
Ketotifen
Imatinib
Fimepinostat
Everolimus
DMSO
Digoxin
Copanlisib
Cabozantinib
Binimetinib
ARV−825
0.00
0.25
0.50
0.75
1.00
Treatment
Proportion of cells with parent organoid
Tumor type
cNF
pNF
MPNST
Other